

## Computer Studies, Chapter 6 – Object Oriented Concepts

### Exam Oriented MCQs, 6.1 Introduction

Q-1: Which of the following methodology are proclaimed to be more reliable, easier to maintain, reuse and enhance in software development?

- A. Sequential      ~~B. Object-oriented~~      C. Structural      D. Procedural

Q-2: In which year the object-oriented programming concepts started originating?

- A. 1990s      B. 1980s      ~~C. 1960s~~      D. 1970s

Q-3: Which of the following is the popular programming language which support object-oriented?

- A. Java      B. C++      C. VB.net      ~~D. All A, B & C~~

Q-4: In which of the following categories the way of programming can be divided?

- A. Procedural      B. Object-oriented      ~~C. Both A & B~~      D. None of these

Q-5: In which programming the focus is on writing functions or procedures which operate on data?

- A. Systematic      ~~B. Procedural~~      C. Object-oriented      D. 'C' language

Q-6: In which paradigm, the focus is on objects?

- A. Structural      B. Procedural      C. Sequential      ~~D. Object-oriented~~

Q-7: Which of the following contain data and functionality both together?

- A. Database      ~~B. Objects~~      C. Class      D. Method

Q-8: Which kind of programming language enables the programmer to create modular, reusable and extendable code?

- A. Procedural      B. Structural      ~~C. Object-oriented~~      D. Sequential

Q-9: By using which kind of programming language, flexibility is gained by being able to change or replace modules without disturbing other parts of code?

- A. Procedural      B. Higher level      ~~C. Object-oriented~~      D. 'C' programming

Q-10: Which is the fundamental building block used by Object-oriented programming?

- ~~A. Object~~      B. Class      C. Person      D. Book

Q-11: Objects are classified using a concept of...

- A. Person      ~~B. Class~~      C. Book      D. Structure

Q-12: How many object properties are supported by object-oriented language?

- A. One      B. Two      C. Three      ~~D. Four~~

Q-13: With which of the following properties a computer language is known as object oriented?

- A. Abstraction, Encapsulation, Polymorphism, Inheritance  
B. Speed of development, Flexibility, Complexity, Modularity  
C. Abstraction, Reusability, Flexibility, Inheritance  
D. Encapsulation, Polymorphism, Reusability, Complexity

## 6.2 Object

Q-14: In Object-oriented terminology, characteristics are known as...

- A. Class      B. Object      ~~C. Properties~~      D. Behaviour

Q-15: Which of the following have unique identity and are distinguishable from each other?

- A. Class      B. Abstraction      ~~C. Object~~      D. Properties

Q-16: What is the name of characteristics in Object-oriented terminology?

- ~~A. Attributes~~      B. Class      C. Method      D. Behaviour

Q-17: What is the other name of Attributes?

- A. Method      ~~B. Properties~~      C. State      D. Value

Q-18: Which values are used to identify the objects?

- A. State      B. Class      ~~C. Attribute~~      D. Member

Q-19: Which term is used for the value of attributes, used to identify the objects?

- A. Method      B. Class      ~~C. State~~      D. Features

Q-20: What is the other name of 'behaviour'?

- A. State      ~~B. Method~~      C. Properties      D. Attributes

Q-21: Which of the following can be changed due to object's behaviour?

- A. Class      ~~B. State~~      C. Method      D. Members

Q-22: Which of the following describes the object?

- A. Members      ~~B. Attributes~~      C. Methods      D. Features

Q-23: Which of the following are data attributes and behavioural methods associated with an object?

- A. Members      B. Features      ~~C. Both A & B~~      D. Method

Q-24: In object-oriented programming, attributes that describe the object are also referred to as...

- ~~A. Data fields~~      B. Properties      C. Features      D. State

### 6.3 Class

Q-25: Which of the following concept enables to express the set of objects that are abstractly equivalent, differing only in the values of their attributes?

- A. Members      ~~B. Class~~      C. Behaviour      D. Features

Q-26: Which of the following can be considered as a blueprint for various objects?

- A. State      B. Attribute      C. Features      ~~D. Class~~

Q-27: Which of the following is a template for multiple objects with similar features?

- ~~A. Class~~      B. Method      C. Feature      D. Properties

Q-28: Which of the following describes a group of objects with similar attributes and common behaviour?

- A. Properties      B. Members      ~~C. Class~~      D. Features

Q-29: Which of the following is a general concept used to embody all the common features of a particular set of objects?

- ~~A. Class~~      B. State      C. Behaviour      D. Attribute

### 6.4 Introduction to Class Diagram

Q-30: Which of the following presents a collection of classes, constraints and relationship among classes?

- A. State diagram      B. Object class      C. Behaviour of class      ~~D. Class diagram~~

Q-31: What is the full form of UML?

- A. Universal Modelling Language      B. Universal Modified Language      ~~C. Unified Modelling Language~~      D. Unified Model Location

Q-32: The full form of OMG is...

- A. Object Management Group
- B. Object Modelling Group
- C. Original Management Group
- D. Object Model Group

Q-33: Which of the following specifies several diagrams for representing different aspects of a software application?

- A. UML
- B. OMG
- C. Methods
- D. Properties

Q-34: Unified Modelling Language is defined by whom?

- A. IMG
- B. PMG
- C. OMG
- D. CMG

Q-35: Which symbol is used to represent class in class diagram?

- A. Circle
- B. Oval
- C. Rectangle
- D. Square

Q-36: In how many sections the rectangle class icon is divided?

- A. One
- B. Two
- C. Three
- D. Four

Q-37: Which of the following are sections of class icon?

- A. Name of the class
- B. Attributes or Properties
- C. Behaviour or operations or methods
- D. All A, B & C

Q-38: Which of the following is in the top section of class icon?

- A. Name of the class
- B. Attributes of the class
- C. Properties of the class
- D. Behaviour of the class

Q-39: Which of the following is in the middle section of class icon?

- A. Methods of the class
- B. Name of the class
- C. Operations of the class
- D. Attributes of the class

Q-40: Which of the following is in the bottom section of class icon?

- A. Behaviour of the class
- B. Operations of the class
- C. Methods of the class
- D. All A, B & C

Q-41: Which of the following is proper syntax of an attribute?

- A. [<visibility>] <initial value> [:<attribute data type>] [=<initial value>]]
- B. [<visibility>] <attribute name> [:<attribute data type>] [=<initial value>]]
- C. [<attribute name>] <visibility> [:<attribute data type>] [=<initial value>]]
- D. [<visibility>] <attribute name> [:<initial value>] [=<attribute data type>]]

Q-42: In syntax of attribute, which values are optional?

- A. Items written within [] brackets
- B. Items written within <> brackets
- C. Both A & B
- D. None of these

Q-43: Which values are supposed to specify by user, in class diagram?

- A. For items enclosed within angular brackets <>
- B. For items within square brackets
- C. For items within parenthesis
- D. Any A, B & C

Q-44: Which kind of visibilities are possible?

- |                                             |           |                   |
|---------------------------------------------|-----------|-------------------|
| A. Private, protected, public or package    | <b>OR</b> | A. -, #, + and ~  |
| B. Class, behaviour, method or attribute    |           | B. -, #, + and \$ |
| C. Method, protected, package or public     |           | C. -, #, + and _  |
| D. Attributes, public, behaviour or private |           | D. +, /, n and *  |

Q-45: Which of the following symbol is used to represent "Private" visibility?

- |      |      |      |      |
|------|------|------|------|
| A. # | B. - | C. + | D. ~ |
|------|------|------|------|

Q-46: Which of the following symbol is used to represent "Protected" visibility?

- |      |      |      |      |
|------|------|------|------|
| A. # | B. ~ | C. + | D. - |
|------|------|------|------|

Q-47: Which of the following symbol is used to represent "Public" visibility?

- |      |      |      |      |
|------|------|------|------|
| A. ~ | B. # | C. - | D. + |
|------|------|------|------|

Q-48: Which of the following symbol is used to represent "Package" visibility?

- |      |      |      |      |
|------|------|------|------|
| A. + | B. # | C. ~ | D. - |
|------|------|------|------|

Q-49: Which of the following is/are example/s of declaring attributes?

- |                  |                            |               |                   |
|------------------|----------------------------|---------------|-------------------|
| A. name : string | B. - balance : float = 0.0 | C. Both A & B | D. person -> name |
|------------------|----------------------------|---------------|-------------------|

Q-50: Which of the following members of the class are not available directly to outside world?

- A. Pure                      ~~B. Private~~                      C. Public                      D. Position

Q-51: Which diagrams are independent of the programming language used for coding an application?

- A. Class                      B. Method                      ~~C. UML~~                      D. Object

## 6.5 Encapsulation

Q-52: Which of the following are core elements for any computer program?

- A. Data and Attributes                      B. Methods and Attributes  
C. ~~Data and Function~~                      D. Behaviour and states

Q-53: In which kind of programming, data can be altered by any component of the program?

- A. Object-oriented                      ~~B. Procedural~~                      C. Conceptual                      D. Multilevel

Q-54: Identify: "The mechanism of providing protection to data and methods of a program." **OR** In object-oriented programming, which of the following property keeps the data safe from unintended actions and inadvertent access by outside objects?

- A. ~~Encapsulation~~                      B. Abstraction                      C. Polymorphism                      D. Hiding

Q-55: What is possible by wrapping data and methods into a single unit class and declaring them as private?

- A. Abstraction                      ~~B. Encapsulation~~                      C. Polymorphism                      D. Messaging

Q-56: Which of the following capability is provided by encapsulation?

- A. Data Abstraction                      B. Polymorphism                      ~~C. Data Hiding~~                      D. Single entity

Q-57: Which kind of members of the class are not available directly to outside world?

- ~~A. Private~~                      B. Public                      C. Package                      D. Protected

Q-58: Which of the following can keep the data safe from unintended actions and inadvertent access by outside objects?

- A. Data abstraction                      ~~B. Encapsulation~~                      C. Polymorphism                      D. Aggregation

Q-59: Which programming discourages direct access to common data by other programs?

- A. Conceptual                      B. Multilevel                      C. Procedural                      ~~D. Object-oriented~~

## 6.6 Data Abstraction



Q-60: Which of the following is a process of representing the essential features of the objects without including implementation detail?

- A. Encapsulation                      B. Aggregation                      C. UML diagrams                      ~~D. Data abstraction~~

OR

Which of the following is a concept that hides the complexity; it says what it does, but not how it is done?

- A. Alert                      B. Messaging                      C. Encapsulation                      ~~D. Data abstraction~~

Q-61: Which concept hides the complexity?

- A. UML diagrams                      B. Class diagrams                      ~~C. Abstraction~~                      D. Encapsulation

Q-62: Which concept says what it does not how it is done?

- ~~A. Abstraction~~                      B. Encapsulation                      C. Messaging                      D. Polymorphism

Q-63: What will be provided by a user-defined function with necessary input data parameters?

- A. Encapsulation                      ~~B. Data abstraction~~                      C. Messaging                      D. Polymorphism

Q-64: Which of the following provides the skeleton or templates for our use?

- A. Messaging                      B. Polymorphism                      C. Encapsulation                      ~~D. Data abstraction~~

Q-65: Which technique relies on the separation of interface and implementation?

- A. Aggregation                      B. Encapsulation                      C. Polymorphism                      ~~D. Data Abstraction~~

Q-66: Which are the examples of data abstraction in C/C++?

- A. Abstract Data Types                      B. Structures                      ~~C. Both A & B~~                      D. Classes

Q-67: Which is the example of data abstraction in C++/Java?

- A. ADT                      ~~B. Classes~~                      C. Structures                      D. Messaging

Q-68: What is the full form of ADT?

- A. ~~Abstract Data Types~~                      B. Abstraction of Data Transmission  
C. Aggregation of Data Types                      D. None of these

Q-69: Which of the following protects data by making them inaccessible from outside?

- A. Data abstraction                      ~~B. Encapsulation~~                      C. Inheritance                      D. Polymorphism

Q-70: Which of the following enables to represent data in which the implementation details are hidden (abstracted)?

- A. Data abstraction      B. Encapsulation      C. Inheritance      D. Messaging

## 6.7 Messaging

Q-71: Identify: "A call to a method, in object-oriented terminology."

- A. Data abstraction      B. Encapsulation      C. Message      D. Aggregation

## 6.8 Polymorphism

Q-72: 'Polymorphism' means

- A. Many forms      B. A call to a method      C. Data hiding      D. Encapsulation

Q-73: Which kind of programming allows defining more than one method having same name but different signatures in a single class?

- A. Structures      B. Object-oriented      C. Procedural      D. All A, B & C

Q-74: What is called, the capability of using same names to mean different things in different contexts?

- A. Overloading      B. Overridden      C. Layout      D. Messaging

Q-75: Object oriented programming allows defining more than one method having same name but different signatures in a single class is known as ...

- A. Polymorphism      B. Messaging  
C. Method overloading OR Function Overloading      D. None of these

Q-76: Which kind of overloading is used, when same operation is given different meanings depending upon the data type of operands used?

- A. Operator overloading      B. Function overloading  
C. Method overloading      D. Both B & C

Q-77: How many types of overloading is used to achieve polymorphism?

- A. One      B. Two      C. Three      D. Four

Q-78: Which of the following are types of overloading in polymorphism achievement?

- A. Function overloading      B. Operator overloading      C. Both A & B      D. Data overloading

## 6.9 Aggregation and Composition

Q-79: When objects of one class are composed of objects of other class, it is called...

- A. Messaging or Polymorphism      B. Aggregation or Composition  
C. Encapsulation or Abstraction      D. Polymorphism or Abstraction



Q-80: Between classes, which kind of relationship is represented by aggregation?

- A. Has – a                      B. a-part-of                      ~~C. Both A & B~~                      D. is – a

## 6.10 Aggregation Vs. Composition

Q-81: Which kind of relationship is represented by aggregation between two classes?

- A. Exclusive                      B. is – a                      C. of – a                      ~~D. Non-exclusive~~

Q-82: Which kind of relationship is represented by composition between two classes?

- ~~A. Exclusive~~                      B. is – a                      C. of – a                      D. Non-exclusive

Q-83: Which is a strong type of aggregation where the lifetime of the part class depends on the existence of the owner class?

- A. Aggregation                      B. Polymorphism                      ~~C. Composition~~                      D. Inheritance

Q-84: With which symbol, basic aggregation is represented?

- A. Filled diamond                      ~~B. Empty diamond~~                      C. Filled rectangle                      D. Empty rectangle

Q-85: Which of the following represents exclusive relationship between two classes?

- A. Aggregation                      B. Polymorphism                      C. Inheritance                      ~~D. Composition~~

Q-86: Which of the following is a strong type of aggregation where the lifetime of the part class is not depend on the existence of the owner class?

- A. Inheritance                      B. Aggregation                      ~~C. Composition~~                      D. Polymorphism

## 6.11 Inheritance

Q-87: Which of the following is the capability of defining a new class of objects that inherits the characteristics of another existing class?

- A. Aggregation                      ~~B. Inheritance~~                      C. Polymorphism                      D. Encapsulation

Q-88: Which other names are available for “New class”, in object-oriented terminology?

- A. Sub class                      B. Child class                      C. Derived class                      ~~D. All A, B & C~~

Q-89: Which other names are available for “Existing class”, in object-oriented terminology?

- A. Super class                      B. Parent class                      C. Base class                      ~~D. All A, B & C~~

Q-90: In class diagram, which symbol is used to represent inheritance?

- A. An arrow pointing to subclass
- B. Two-sided arrow which connects subclass and superclass
- C. An arrow pointing to superclass
- D. A single line connecting sub and superclass

Q-91: Teacher is a kind of person is an example of which of the following options?

- A. Aggregation
- B. Inheritance
- C. Data Abstraction
- D. Encapsulation

Q-92: Which of the following example represents multilevel inheritance?

- A. date1 – date2 and n1 – n2
- B. Motherboard is a part of computer
- C. Employee is a kind of person and teacher is an employee
- D. Call to a method

Q-93: Which of the following is another name for inheritance or “is - a” relationship?

- A. Polymorphism
- B. Generalization
- C. Aggregation
- D. Encapsulation

Q-94: Which of the following refers to a relationship between two classes where one class is a specialized version of another?

- A. Generalization
- B. Composition
- C. Aggregation
- D. Inheritance

Q-95: Name the inheritance, where there may be a classical hierarchy of inheritance between class.

- A. Multiple inheritance
- B. Multilevel inheritance
- C. Conceptual inheritance
- D. Complex inheritance

Q-96: When a class is derived from two or more classes, it is known as...

- A. Polymorphism
- B. Multiple inheritance
- C. Generalization
- D. Aggregation

## 6.12 Composition Vs. Inheritance

Q-97: In which of the following, class inherits from other classes in order to share, reuse or extend functionality?

- A. Inheritance
- B. Composition
- C. Aggregation
- D. Encapsulation

Q-98: Which kind of relationship is existed between superclass and subclass in inheritance?

- A. Has – a
- B. this – is – a
- C. Has – below
- D. A kind of

Q-99: In which of the following classes do not inherit from other classes, but are ‘composed of’ other classes?

- A. Inheritance
- B. Polymorphism
- C. Composition
- D. Encapsulation

Q-100: In which of the following, class contains the attributes where some attributes are of objects of other class types?

- A. Inheritance
- B. Composition
- C. Polymorphism
- D. Encapsulation